SMART CITY DATA ANALYSIS

DELHI TRAFFIC DATA ANALYSIS USING PYTHON

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DATE : 08-APRIL -2026

INTRODUCTION

Smart cities use technology and data to improve urban life.

- Traffic analysis helps in:
- Reducing congestion
- Improving road safety
- Better traffic management
- Smart transportation planning

This project analyzes Delhi traffic data using Python.

OBECTIVE

Objectives of the project:

i. Analyze traffic patterns in Delhi
ii. Study traffic density levels
iii. Understand speed variation
iv. Visualize traffic data using graphs

DATASET INFORMATION

Dataset contains the following features:

- Trip ID
- Start Area
- End Area
- Distance (km)
- Time of Day
- Day of Week
- Weather Condition
- Traffic Density Level
- Average Speed (kmph)

TECHNOLOGIES USED

Tools used in this project:

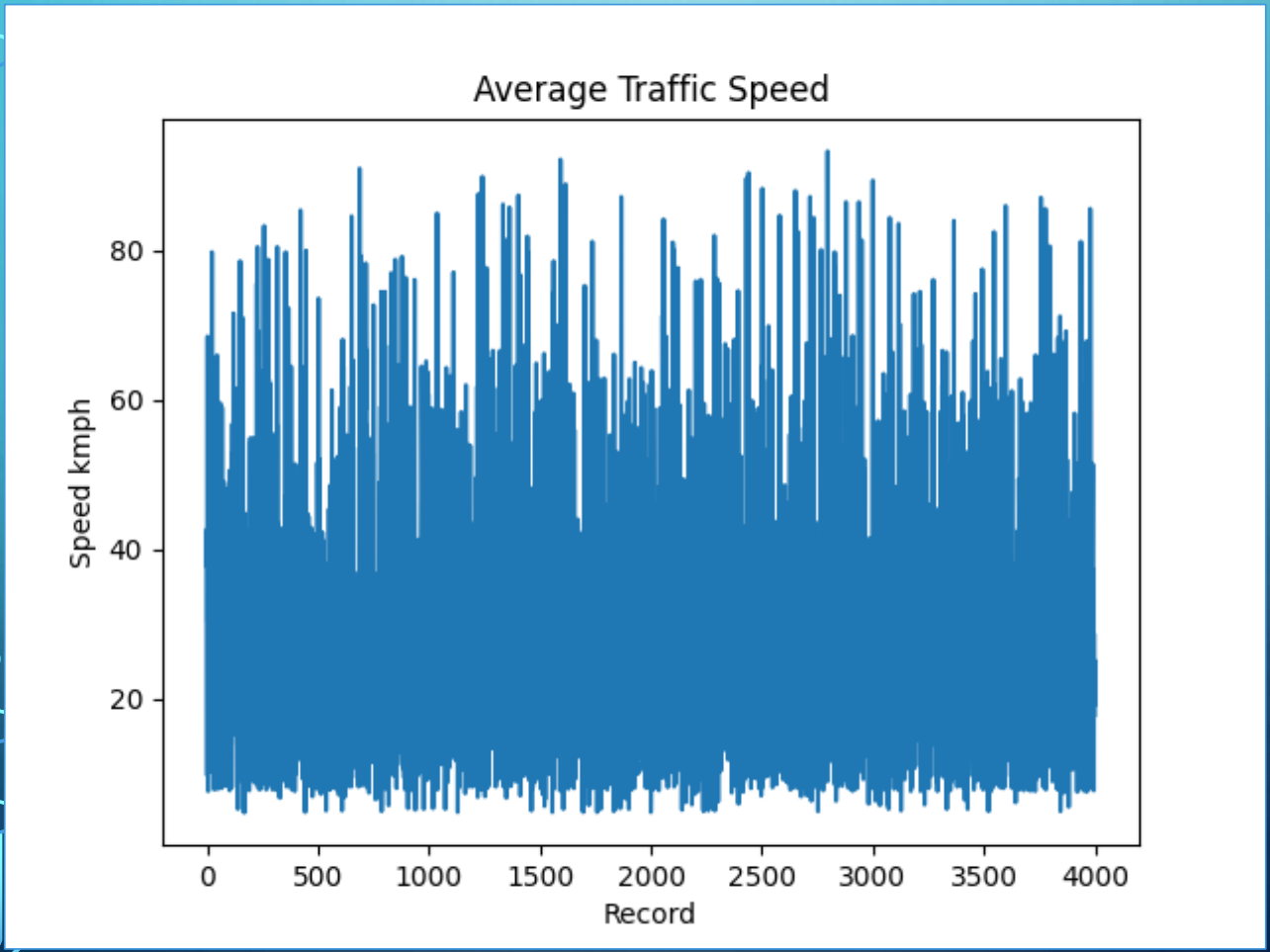
- Python
- Pandas
- Matplotlib
- Google Colab
- Power BI (for dashboard visualization)

DATA PREPROCESSING

Steps performed:

- Loading CSV dataset
- Cleaning missing values
- Checking data types
- Filtering useful attributes

AVERAGE TRAFFIC SPEED ANALYSIS



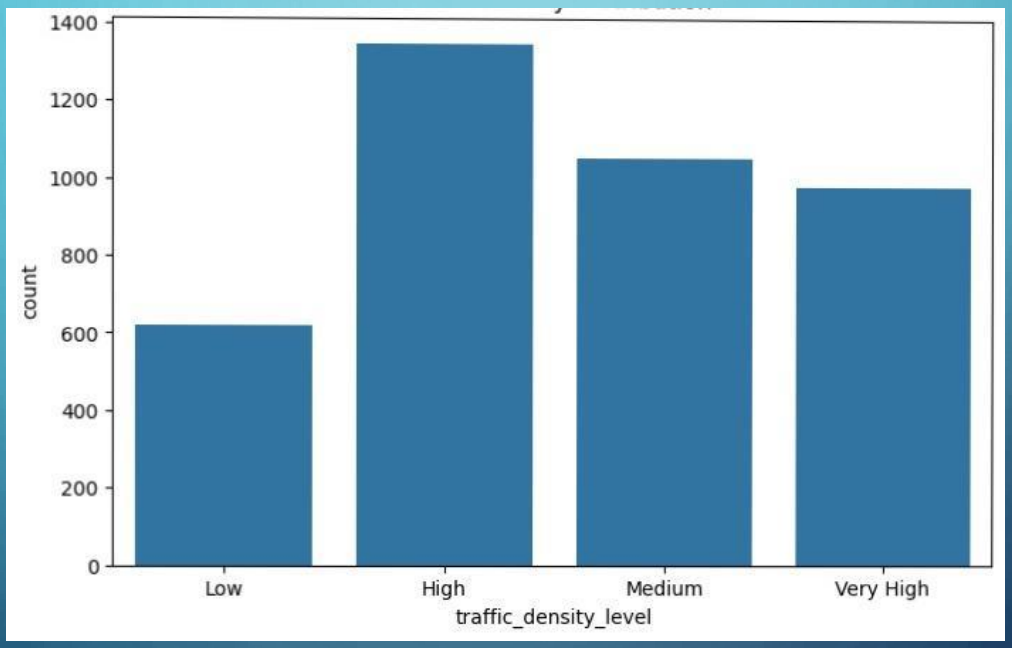
TRAFFIC DENSITY ANALYSIS

Explain levels:

- Low traffic
- Medium Traffic
- High Traffic

Graph showing traffic density distribution.

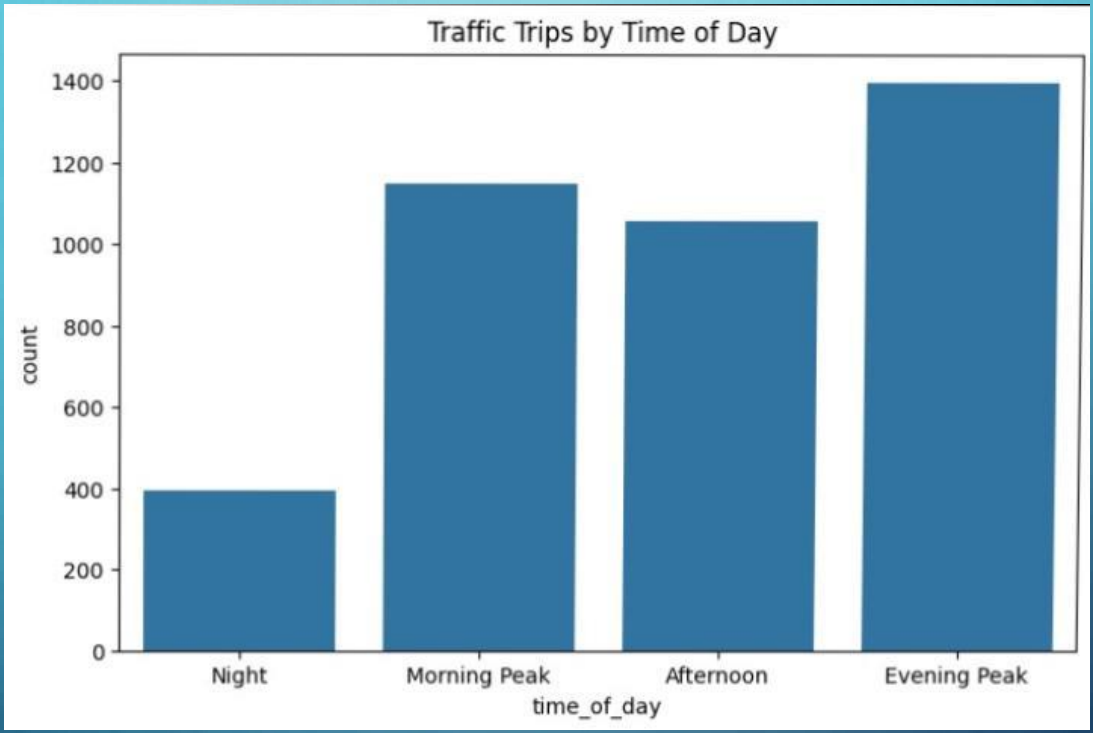
TRAFFIC DENSITY DISTRIBUTION



TIME OF DAY ANALYSIS

Traffic varies during:

- Morning peak hours
- Afternoon moderate traffic
- Evening peak congestion


Traffic Trips by Time of Day
1400
1200
1000
800
600
400
200
0
count
Night
Morning Peak
Afternoon
Evening Peak
time_of_day

CONCLUSION

From the analysis:

- Traffic speed decreases during peak hours
- High density areas show heavy congestion
- Data visualization helps in better decision making